

1. 2012Q15

Identical twins have the same

- (1) genotype.
 - (2) traits involving continuous variations.
 - (3) traits involving discontinuous variations.
- A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

2. 2012Q18

Which of the following descriptions about transcription is correct?

- A. tRNA pairs with ribosome to produce amino acids.
- B. mRNA pairs with ribosome to produce proteins.
- C. Free DNA nucleotides pair with DNA template strand to produce DNA.
- D. Free RNA nucleotides pair with DNA template strand to produce mRNA.

3. 2012Q19

The tRNA anticodon for the sequence AGT on the coding strand of DNA is

- A. UCA.
- B. AGU.
- C. TCA.
- D. AGT.

4. 2013Q4

For different DNA nucleotides, the part that varies from one to another is the

- A. base.
- B. ribose.
- C. deoxyribose.
- D. phosphate group.

5. 2013Q11

Which of the following statements about codons is correct?

- A. A codon may consist of bases A, C, G, or T.
- B. A codon is a triplet of bases on transfer RNA.
- C. Most amino acids are coded by more than one codon.
- D. All codons code for amino acids.

6. 2013Q12

The DNA of a eukaryotic cell contains 20% adenine (A) bases. What is the percentage of cytosine (C) in this DNA?

- A. 60%
- B. 40%
- C. 30%
- D. 20%

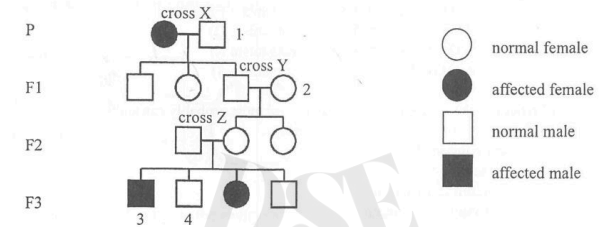
7. 2013Q13

A man and his wife are heterozygous for blood type A and B respectively. What is the probability of their son having blood type AB?

- A. 0
- B. 1/4
- C. 1/2
- D. 3/4

8. 2013Q15

Questions 15 to 17 refer to the pedigree below, which shows the inheritance of a certain trait controlled by a pair of alleles located on an autosome (i.e non-sex chromosome):



Which cross(es) can be used to deduce which phenotype is dominant?

- A. cross Y only
- B. cross Z only
- C. crosses X and Y only
- D. crosses X and Z only

9. 2013Q16

What are the probable genotypes of individuals 1 and 2?

- | | Individual 1 | Individual 2 |
|----|----------------------------|----------------------------|
| A. | homozygous | homozygous |
| B. | homozygous | homozygous or heterozygous |
| C. | homozygous or heterozygous | homozygous |
| D. | homozygous or heterozygous | homozygous or heterozygous |

10. 2013Q17

If individuals 3 and 4 are twins, which of the following conclusions can be drawn?

- A. They are developed from the same fertilised egg.
- B. They are developed from different fertilised eggs.
- C. They are genetically different for characters displaying continuous variation.
- D. They are genetically different for characters displaying discontinuous variation.

11. 2013Q19

Question 19 refers to the nucleotide sequence of a certain functional gene segment found in four different species of organisms P, Q, R and S:

A A C G T C G A A A (organism P)
 A A C C T C G A A A (organism Q)
 A G G C T A G A A A (organism R)
 A G G C T A G T A A (organism S)

The differences in the sequences shown above are most probably caused by

- A. crossing over.
- B. gene mutation.
- C. random fertilization.
- D. chromosomal mutation.

12. 2014Q10

Questions 10 and 11 refer to the following two crosses of fruit flies. In fruit flies, males are the heterogametic sex (XY) and the wing shape (normal wing or cut wing) is controlled by a single gene.

	<i>Cross I</i>	<i>Cross II</i>
Parents	Normal wing female x Cut wing male	Cut wing female x Normal wing male
F ₁	12 normal wing females 11 normal wing males	11 normal wing females 11 cut wing males
F ₂	71 normal wing females 34 normal wing males 35 cut wing males	32 normal wing females 33 cut wing females 36 normal wing males 38 cut wing males

Which of the following observations from Cross I best supports the conclusion that normal wing is the dominant phenotype?

- A. All the F₁ individuals are normal wing.
- B. The ratio of normal wing individuals to cut wing individuals in F₂ is 3:1.
- C. There are more normal wing individuals than cut wing individuals in the F₂.
- D. Normal wing males are more or less the same in number as cut wing males in the F₂.

13. 2014Q11

From Cross II, we can conclude that

- A. the gene for the wing shape is located on the X-chromosome because the cut wing phenotype was passed from the female parent to the F₁ males.
- B. the law of independent assortment was demonstrated because new phenotypes, including normal wing females and cut wing males, were found in F₂.
- C. the normal wing male parent is heterozygous because four combinations of phenotypes were observed in F₂.
- D. the gene for the wing shape is located on an autosome because a ratio of 1:1:1:1 was shown in F₂.

14. 2014Q12

Which of the following combinations of blood groups of parents may produce offspring with blood group AB?

- (1) A x B
- (2) AB x O
- (3) AB x AB
- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

15. 2014Q16

Which of the following biomolecules are associated with transcription?

- (1) DNA
- (2) mRNA
- (3) amino acid
- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

16. 2015Q16

Variation in skin colour exists among different human races. Which of the following factors plays the major role in determining this variation?

- A. exercise
- B. nutrition
- C. inheritance
- D. exposure to sunlight

17. 2015Q26

During pregnancy, amniotic fluid containing foetal cells can be obtained for karyotyping. This helps to determine whether the foetus

- (1) is male or female.
- (2) has Down syndrome or not.
- (3) is a carrier of Sickle-cell anaemia.
- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

18. 2016Q4

Which of the following parts of the nucleotide make up the backbone of a polynucleotide strand?

- A. sugar and base
- B. sugar and phosphate
- C. base and phosphate
- D. base, sugar and phosphate

19. 2017Q8

In humans, blood group B is dominant to blood group O. In a family, the father and mother are of blood groups O and B respectively. They have two children who are of blood group B. The father concludes that his wife must be homozygous for blood group B. Is this conclusion correct?

- A. No, because there are other blood groups besides blood groups B and O.
- B. No, because even if the mother is heterozygous, each child has a 50% chance to be of blood group B.
- C. Yes, because the father has no allele for blood group B, all alleles for blood group B must have come from the mother.
- D. Yes, because if the mother is heterozygous, one child should be of blood group B and the other should be of blood group O.

20. 2017Q9

If the base sequence on the coding strand of the DNA is AAC, which of the following combinations correctly shows the mRNA codon and the tRNA anticodon?

- | | mRNA codon | tRNA anticodon |
|----|-------------------|-----------------------|
| A. | AAC | UUG |
| B. | AAC | TTG |
| C. | UUG | AAC |
| D. | TTG | AAC |

21. 2018Q12

Questions 12 and 13 refer to the information below. Kathy had two pure-bred cats, one had long white fur while the other had short black fur. It is known that fur length and fur colour are controlled by two different genes respectively. The two cats gave birth to four kittens which had long black fur.

Which of the following conclusions can be drawn based on the above case?

- (1) Long fur is dominant over short fur.
 - (2) The four kittens have the same genotype for fur length and fur colour.
 - (3) The genes controlling fur length and fur colour are located on different chromosomes.
- A. (1) and (2) only
 - B. (1) and (3) only
 - C. (2) and (3) only
 - D. (1), (2) and (3)

22. 2018Q13

After the kittens had grown up, they interbred and gave birth to the second filial generation (F_2). Among the F_2 kittens, there was one with short white fur. Which of the following processes most likely contributed to the occurrence of this new phenotype?

- (1) Mutation
 - (2) Random fertilisation of gametes
 - (3) Independent assortment of chromosomes
- A. (1) and (2) only
 - B. (1) and (3) only
 - C. (2) and (3) only
 - D. (1), (2) and (3)

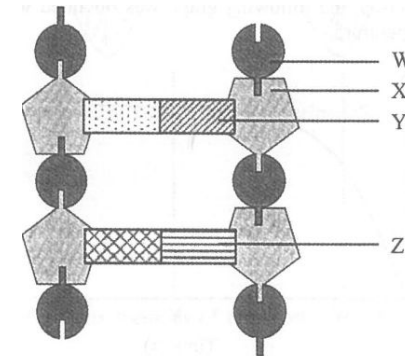
23. 2018Q28

The DNA model proposed by Watson and Crick leads to the understanding of how

- A. organisms store genetic codes.
- B. organisms share a common ancestor.
- C. cells produce instructions for protein synthesis.
- D. cells can pass genetic information to the next generation.

24. 2019Q13

The diagram below shows a DNA model:



Which of the following combinations shows the most probable identities of molecules W, X, Y, and Z?

- | | W | X | Y | Z |
|----|-----------|-----------|----------|----------|
| A. | sugar | phosphate | cytosine | thymine |
| B. | sugar | phosphate | cytosine | guanine |
| C. | phosphate | sugar | adenine | thymine |
| D. | phosphate | sugar | adenine | guanine |

25. 2019Q14

In a family, the father is red-green colour blind (a recessive X-linked trait) and is of blood group A, while the mother has normal vision and is of blood group B. Which of the following could be the phenotypes of their biological child?

- (1) A normal-vision girl with blood group O
 - (2) A red-green colour blind girl with blood group O
 - (3) A red-green colour blind boy with blood group AB
- A. (1) and (2) only
 - B. (1) and (3) only
 - C. (2) and (3) only
 - D. (1), (2) and (3)

26. 2020Q18

Questions 18 and 19 refer to the following passage about the discovery of DNA structure:

Many scientists tried to uncover the structure of DNA. In 1952, Franklin took the first X-ray photograph of DNA, which revealed its helical shape. Her colleague, Wilkins, showed some of Franklin's unpublished findings to Watson without her knowledge. Shortly after, Watson and Crick made a crucial advance when they proposed that the DNA molecule was made up of two chains of nucleotides paired to form a double helix. In 1962, Watson, Crick, and Wilkins were awarded the Nobel Prize.

Which of the following aspects of the Nature of Science can be exemplified in the above story?

- (1) A good scientific experiment must include carefully designed controls.
 - (2) Scientists are both collaborative and competitive by nature.
 - (3) Doing science requires imagination and creativity.
- A. (1) and (2) only
 - B. (1) and (3) only
 - C. (2) and (3) only
 - D. (1), (2) and (3)

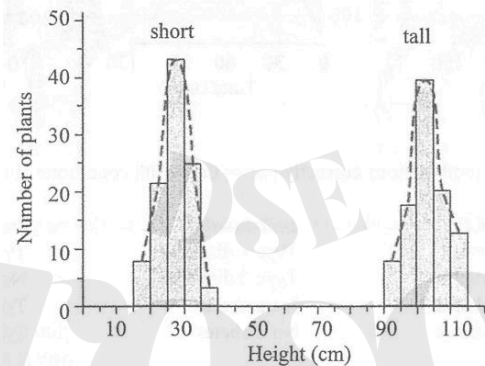
27. 2020Q19

Which of the following descriptions of the molecular structure of DNA proposed by Watson and Crick is correct?

- A. The two chains run in opposite directions.
- B. The bases link up the nucleotides to form a chain.
- C. The sugar that makes up nucleotides has two types.
- D. The phosphate that makes up nucleotides has at least four types.

28. 2020Q33

Questions 33 and 34 refer to the following graph, which shows the variations in the height of a certain type of plant:



Which of the following conclusions can be drawn from the above graph?

- A. The short and tall plants are of two different species.
- B. The two traits, short and tall, are controlled by a pair of alleles.
- C. More samples should be taken to cover the full range of heights.
- D. The height of the plants displays the properties of continuous and discontinuous variations.

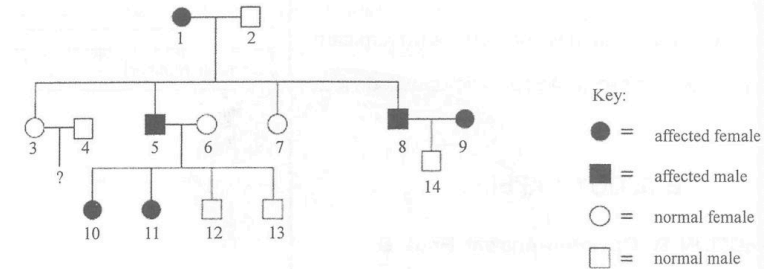
29. 2020Q34

Which of the following factors contributes *least* to the variations shown?

- A. the height of the parental plants
- B. the light intensity in the environment
- C. the independent assortment of chromosomes
- D. the oxygen concentration in the environment

30. 2020Q35

Questions 35 and 36 refer to the pedigree below, which shows the inheritance of a genetic disorder.



Which of the following combinations correctly shows the possible types of inheritance of this disorder?

	<i>Autosomal dominant</i>	<i>Autosomal recessive</i>	<i>X-linked dominant</i>	<i>X-linked recessive</i>
A.	✓		✓	
B.	✓			✓
C.		✓	✓	
D.		✓		✓

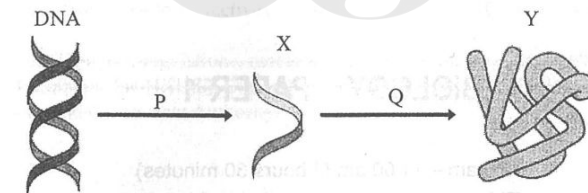
31. 2020Q36

What is the probability of individual 3 and individual 4 giving birth to a normal male?

- A. 0
- B. 0.25
- C. 0.5
- D. 1

32. 2021Q1

Questions 1 to 3 refer to the diagram below, which shows the flow of gene expression within a cell:



X represents

- A. a transfer RNA.
- B. a DNA template.
- C. a ribosomal RNA.
- D. a messenger RNA.

33. 2021Q2

Y could be

- (1) an enzyme.
- (2) a hormone.
- (3) an antibody.
- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

34. 2021Q3

Which of the following combinations correctly states what process P is and where process Q takes place?

	What process P is	Where process Q takes place
A.	translation	cytoplasm
B.	translation	nucleus
C.	transcription	cytoplasm
D.	transcription	nucleus

35. 2021Q11

The human ABO blood group system is controlled by multiple alleles. Alleles I^A and I^B lead to the presence of antigens A and B on the surface of red blood cells respectively. Allele i leads to the absence of antigens A and B on the surface of red blood cells. How many genotypes are involved in the ABO blood group system?

- A. 3
- B. 4
- C. 6
- D. 8

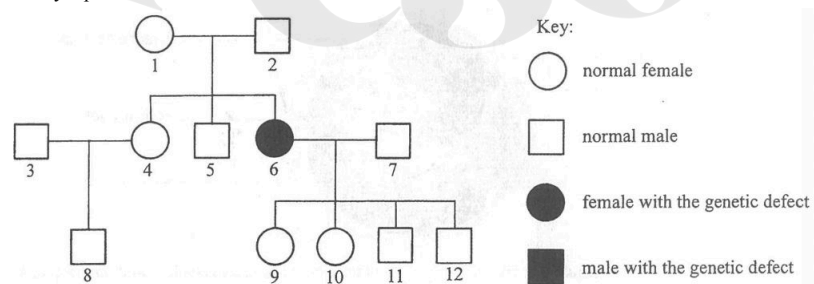
36. 2021Q12

A pair of identical twins were adopted by two different families. After the twins grow up, which of the following descriptions about them will most likely apply?

- A. They have different shapes of earlobe.
- B. They have the same pattern of fingerprint.
- C. They have different degrees of short-sightedness.
- D. They have the same number of freckles on their faces.

37. 2021Q13

Questions 13 to 15 refer to the pedigree below, which shows the inheritance of a certain genetic defect controlled by a pair of alleles.



The phenotype of individual 6 can be explained by the fact that this individual received a defective allele on

- A. an autosome from each parent.
- B. the Y chromosome from the father.
- C. an X chromosome from the mother.
- D. an X chromosome from each parent.

38. 2021Q14

The offspring of individuals 6 and 7 are normal because each receives a

- A. normal allele from the father.
- B. normal allele from each parent.
- C. defective allele from the father.
- D. defective allele from the mother.

39. 2021Q15

Which of the following correctly shows the possible genotype(s) of individual 8?

- A. heterozygous only
- B. homozygous dominant only
- C. homozygous recessive only
- D. homozygous dominant and heterozygous

40. 2021Q16

In the crossing of a red-flowered plant with a white-flowered plant, half of the F_1 offspring were red-flowered and the other half were white-flowered. When the F_1 white-flowered plants were self-crossed, all the F_2 offspring were white-flowered. What can be concluded?

- A. The red-flowered parental plant was homozygous.
- B. The white-flowered parental plant was heterozygous.
- C. White flower is the dominant character in this species of plant.
- D. Red flower is the dominant character in this species of plant.

41. 2021Q17

Although Mendel proposed that the characteristics of the pea plants were controlled by a pair of "heredity factors", it was not recognised during his lifetime. His idea was later accepted in the early 20th century because it was supported by:

- A. the theory of natural selection.
- B. the same patterns of inheritance in fruit flies.
- C. the behaviours of chromosomes in cell division.
- D. the discovery of the double helix structure of DNA.

42. 2022Q2

Adenine (A) and cytosine (C) together make up 50% of the total amount of nitrogenous bases in DNA molecules. Which of the following features of DNA provide the basis to explain this phenomenon?

- (1) DNA has a helical structure.
- (2) DNA is a double-stranded molecule.
- (3) The nitrogenous bases are paired complementarily.
- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

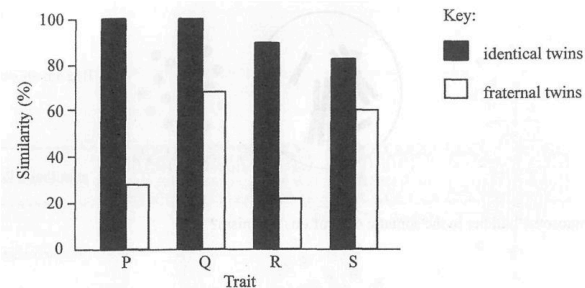
43. 2022Q3

In Mendel's experiment, he proposed that a characteristic of the pea plants was controlled by a pair of "heredity factors." These factors can be separated in a "process" before passing to the offspring. According to the current understanding of genetics, what do the "heredity factors" and the "process" represent respectively?

Heredity factors	Process
A. alleles	first meiotic cell division
B. alleles	second meiotic cell division
C. chromosomes	first meiotic cell division
D. chromosomes	second meiotic cell division

44. 2022Q6

A large survey about the inheritance of four traits in identical twins and fraternal twins has been conducted. The similarity of these four traits among the individuals in each type of twins are shown in the graph below:



Which trait is most affected by environmental factors?

- A. P
- B. Q
- C. R
- D. S

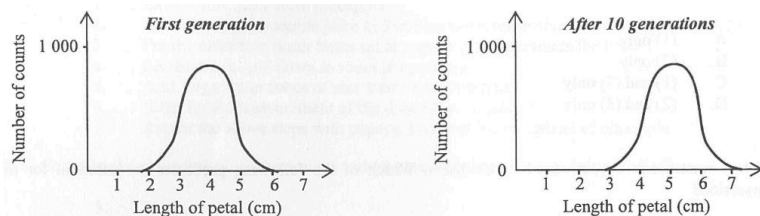
45. 2023Q4

Which of the following processes **does not** involve membrane proteins?

- A. transcription of DNA to form mRNA
- B. absorption of minerals by root hair cells
- C. digestion of carbohydrates in the small intestine
- D. transmission of nerve impulse between two neurones

46. 2023Q24

Question 24 refers to the information about the introduction of a foreign plant species to a local botanic garden. Over the years, the gardener noted that there was a change in the size of its flowers. The graphs below show the variations of the petal length of this plant species in the first generation and after 10 generations.

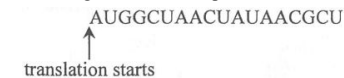


Which of the following correctly describe the inheritance of the petal length in this plant species?

- (1) The petal length is affected by environmental factors.
 - (2) The inheritance of the petal length is controlled by a number of genes.
 - (3) The variation in the petal length is a result of independent assortment.
- A. (1) and (2) only
 - B. (1) and (3) only
 - C. (2) and (3) only
 - D. (1), (2) and (3)

47. 2023Q27

Questions 27 and 28 refer to the following nucleotide sequence found on a mRNA molecule:



How many different codons are found on this nucleotide sequence?

- A. 3
- B. 4
- C. 5
- D. 6

48. 2023Q28

Which of the following statements about the mRNA molecule are correct?

- (1) It can be found in the nucleus.
 - (2) It can be found in the cytoplasm.
 - (3) Its corresponding sequence on the DNA coding strand is TACCGATTGATATTGCGA
- A. (1) and (2) only
 - B. (1) and (3) only
 - C. (2) and (3) only
 - D. (1), (2) and (3)

49. 2023Q30

Question 30 refers to the phylogenetic relationship of five bacterial species. Enzyme X is found in most bacteria. The table below shows the number of nucleotide differences in the genes encoding enzyme X of five bacterial species (J, K, L, M and N):

	J	K	L	M	N
J	0	61	32	112	89
K	-	0	50	119	94
L	-	-	0	119	77
M	-	-	-	0	124
N	-	-	-	-	0

The nucleotide sequence of the genes encoding enzyme X have some differences among the five species. Yet, the enzyme X in these five species still serve the same function. Which of the following are possible explanations for this phenomenon?

- (1) The polypeptides formed have the same length.
 - (2) There are more than one genetic code for the same amino acid.
 - (3) The differences do not affect the shape of the active site of enzyme X.
- A. (1) and (2) only
 - B. (1) and (3) only
 - C. (2) and (3) only
 - D. (1), (2) and (3)

50. 2024Q2

The genetic material of a virus contains 1966 adenine (A), 2343 guanine (G), 1749 thymine (T), and 2368 cytosine (C). Based on this information, which of the following combinations of genetic material for this virus is correct?

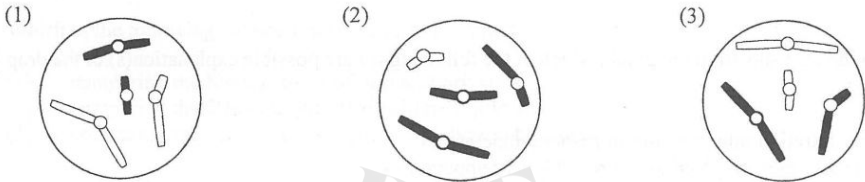
- | | Nucleic acid | Pentose |
|----|---------------------|----------------|
| A. | single-stranded | deoxyribose |
| B. | single-stranded | ribose |
| C. | double-stranded | deoxyribose |
| D. | double-stranded | ribose |

51. 2024Q5

Questions 5 and 6 refer to the diagram below, which shows the appearance of chromosomes in a cell at the early stage of meiotic cell division:



Which of the following diagrams show the possible combinations of the chromosomes in the daughter cells?



- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

52. 2024Q6

Mendel’s law can be used to explain the combinations of chromosomes in the daughter cells in Question 5.

Which of the following correctly shows the corresponding Mendel’s law and the stage of meiotic cell division as described by the law?

	<i>Mendel’s law</i>	<i>Stage</i>
A.	Law of segregation	first meiotic cell division
B.	Law of segregation	second meiotic cell division
C.	Law of independent assortment	first meiotic cell division
D.	Law of independent assortment	second meiotic cell division

52. 2024Q16

Questions 16 to 18 refer to the table below, which shows the results of some crosses in one type of plants.

This type of plant produces individuals either bearing male flowers with stamens only or female flowers with carpels only. The colour of the flower is controlled by a single gene.

<i>Cross</i>	<i>Male parent</i>	<i>Female parent</i>	<i>Offspring</i>
1	white flowers	white flowers	all white flowers
2	red flowers	white flowers	all red flowers
3	white flowers	red flowers	all red flowers
4	red flowers	white flowers	some red flowers, some white flowers
5	?	red flowers	some red flowers, some white flowers

It is known that the sex determination of this plant is similar to that of humans. Which of the following combinations correctly describes the allele for white flower in this plant?

	<i>Dominant or recessive</i>	<i>Location of the allele</i>
A.	dominant	autosome
B.	dominant	sex chromosome
C.	recessive	autosome
D.	recessive	sex chromosome

53. 2024Q17

Which of the following combinations is most likely the result from Cross 4?

	<i>No. of individuals with red flowers</i>	<i>No. of individuals with white flowers</i>
A.	26	73
B.	48	46
C.	69	37
D.	71	25

54. 2024Q18

Which of the following statements about Cross 5 is correct?

- A. The female parent could be a homozygote.
- B. The male parent could be a homozygote with red flowers.
- C. The male parent could be a homozygote with white flowers.
- D. The male parent could be a heterozygote with white flowers.

LQ

1. 2012Q8b, c

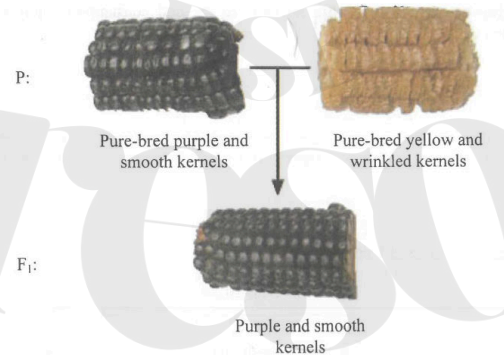
The photograph below shows the appearances of some kernels of a corn:



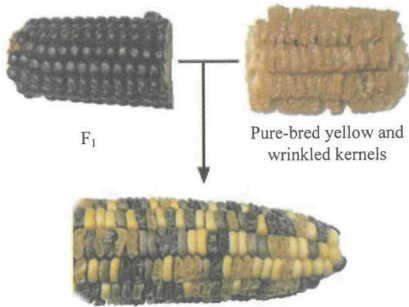
Kernel	Appearance
A	Purple and smooth
B	Purple and wrinkled
C	Yellow and smooth
D	Yellow and wrinkled

The purple colour is produced by a pigmented layer within the kernels. If the layer is not pigmented, the yellow colour of the inner tissue becomes visible. Whether the kernel is smooth or wrinkled is due to the type of food stored inside it. Smooth kernels (starchy corn) store starch while wrinkled kernels (sweet corn) store soluble sugars. The surface of the sweet corn becomes wrinkled when the corn dries up.

- b) The two traits of the kernels are controlled by genes located on different homologous chromosomes. The following diagram shows the result of a cross between two pure-bred corn plants, one with purple and smooth kernels and the other with yellow and wrinkled kernels:



- i) Based on the results of the cross, deduce which phenotypes are dominant. (3 marks)
- ii) The F₁ generation produced was crossed with pure-bred corn plants with yellow and wrinkled kernels, as shown below:

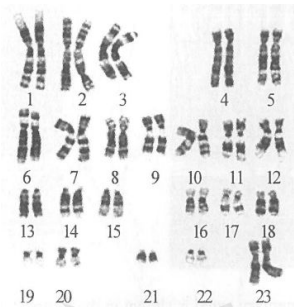


- Explain the results of the cross using Mendel's law of inheritance. (4 marks)
(Marks will not be awarded for genetic diagrams.)
- c) When Mendel proposed how traits are inherited, chromosomes had not yet been discovered. (3 marks)
In your opinion, how did Mendel come up with his hypothesis?

2. 2013Q4

Red-green colour blindness is an X-linked recessive trait in humans. Peter is red-green colour blind while his daughter, Mary, is normal.

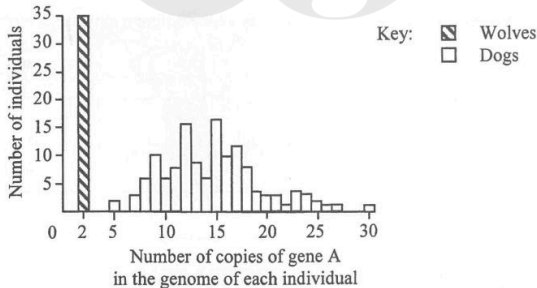
- a) Deduce Mary's genotype without using a genetic diagram. (4 marks)
- b) Mary is an expectant mother. The photomicrograph below shows the karyotype of her foetus:



- i) From the photomicrograph, can we deduce whether this foetus will be red-green colour blind or not? Explain your answer. (2 marks)
- ii) Is the foetus a boy or a girl? Explain your answer with reference to the photomicrograph. (3 marks)

3. 2014Q6a

It is generally believed that domestic dogs evolved from ancient wolves. A recent study comparing the genomes of wolves and dogs suggests that genes with key roles in starch digestion were selected during the domestication of wolves into dogs. One of these genes was gene A, which codes for amylase. This gene may exist in many copies in a genome. The following graph shows the number of individuals having different numbers of copies of gene A in 35 wolves and 136 dogs.

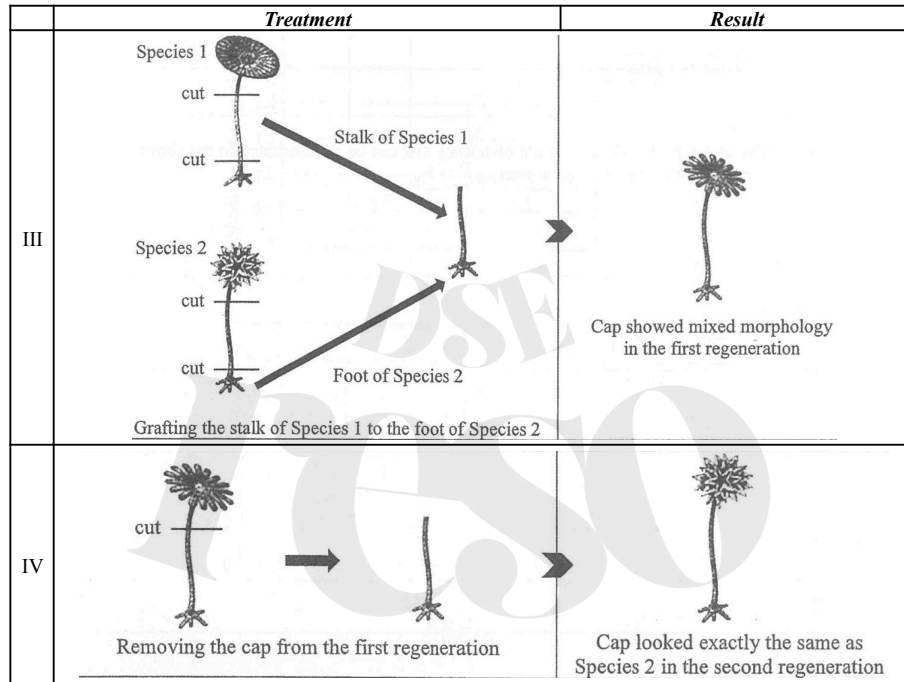


- a) Based on the data above and the gene expression processes, explain why the amylase activity in dogs is generally higher than that in wolves. (3 marks)

- b) Based on the results, Hämmerling hypothesized that:

The hereditary information is stored in the foot of the algal cell.

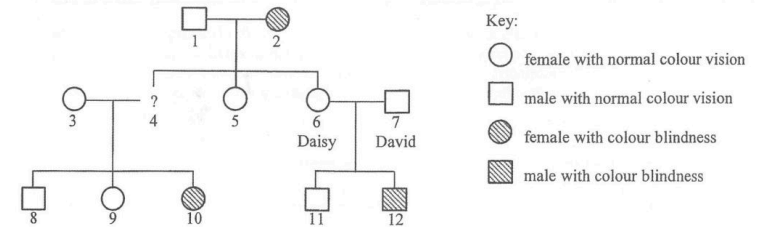
Hämmerling further studied the expression of hereditary information by grafting the stalk of Species 1 to the foot of Species 2. After grafting, the cap formed from the first regeneration showed a mixed morphology (III). He then removed the regenerated cap. The cap formed from the second regeneration looked exactly the same as Species 2 (IV). The diagram below shows the treatments and the results.



- i) Hämmerling concluded that when the cut stalk was transplanted, it contained some short-lived instruction derived from the foot of Species 1, resulting in a mixed morphology of the cap in the first regeneration.
- 1) Suggest the type of biomolecule that carried the short-lived instruction. (1 mark)
 - 2) How could the biomolecule suggested in (1) affect the morphology of the cap? (2 marks)
- ii) How do the results from this experiment support Hämmerling's hypothesis? (2 marks)
- c) Give **one** aspect about the nature of science that can be demonstrated in the above discovery and give a reason to support your answer. (2 marks)

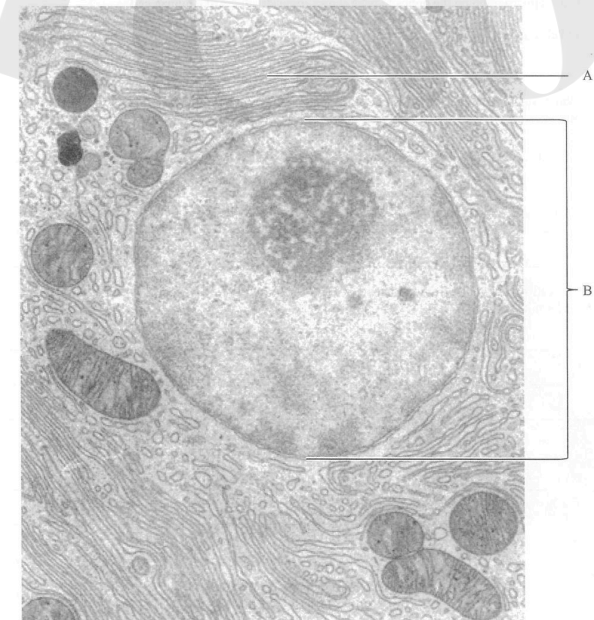
7. 2016Q10b, c, d

Color blindness is an X-linked recessive genetic disorder. The pedigree below shows the inheritance of colour blindness in a family:



- b) Given that the dominant allele of colour vision is represented by R while the recessive allele is represented by r, determine all the possible genotypes and phenotypes of the offspring of individuals 1 and 2 using a genetic diagram. (5 marks)
- c) Draw all possible representations(s) for individual 4 with reference to the key of the pedigree. (1 mark)
- d) Daisy (individual 6) has recently given birth to a baby girl. Since one of her sons suffered from colour blindness (individual 12), Daisy worried that their daughter would have colour blindness too. David (individual 7) reassured her by saying that:
"Don't worry. Our daughter will be fine because I have normal colour vision!"
 Justify David's claim. (5 marks)

8. 2017Q4c



The electron micrograph on the opposite page shows some structures of a human cell.

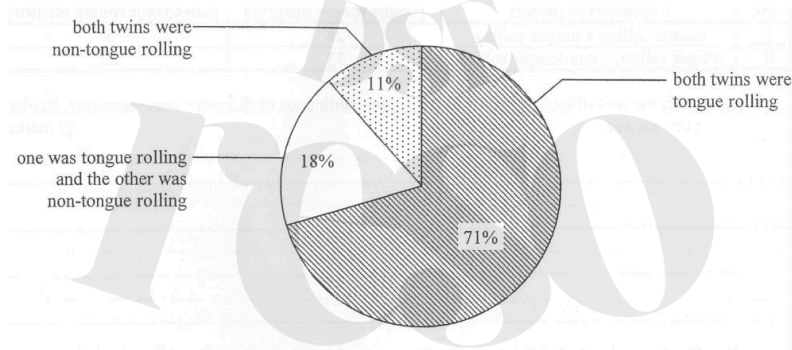
- c) The cell was obtained from the pancreas. How do A and B work together such that this cell can perform its function? (4 marks)

9. 2017Q10

- a) In 1940, scientist Alfred Sturtevant hypothesized that the ability to roll one's tongue is determined by a single gene. His hypothesis was based on the data below:

Case	Characters of parents	Tongue rolling offspring	Non-tongue rolling offspring
I	tongue rolling x tongue rolling	28	5
II	tongue rolling x non-tongue rolling	33	22

- i) Does the trait of tongue rolling ability show continuous or discontinuous variation? (2 marks)
Explain your answer.
- ii) Sturtevant concluded that tongue rolling is the dominant phenotype while non-tongue rolling is the recessive phenotype. With reference to the above table, explain how he arrived at this conclusion. (2 marks)
- b) In 1965, the offspring of a group of non-tongue rolling parents were studied. It was found that more than 30% of the offspring were tongue rollers. Does this finding support Sturtevant's conclusion in (a)(ii)? Explain your answer. (2 marks)
- c) In 1971, another study on identical twins was carried out to further explore the factors influencing the tongue rolling trait. The results are summarised in the chart below:



- i) What is the advantage of using identical twins as the subjects for the study? (2 marks)
- ii) With reference to the chart, complete the following table with data that support the conclusions. (2 marks)

Conclusion	Evidence
Genetic factors play a significant role in the determination of the tongue rolling trait.	
There are other factors influencing the tongue rolling trait.	

- d) i) In the above case regarding the development of knowledge about the inheritance of the tongue rolling trait, which of the following ideas about science is demonstrated? (2 marks)

Ideas about science	Put a '✓' in the appropriate spaces below
Science is a process of ongoing inquiries.	
Science is affected by social and cultural factors.	
Scientists may not arrive at the same conclusions about the same set of data	
Scientists investigations may not require doing experiments in laboratories	

- ii) Elaborate on how the development of knowledge about the inheritance of the tongue rolling trait can be used to demonstrate that scientists have to be open-minded. (1 mark)

10. 2018Q5c

- c) It is commonly thought that the sex of offspring is mainly determined by the mother. (3 marks)
Explain why this is **not** true.

11. 2018Q6a

- a) Of all the blood groups, blood group O is in greatest demand in the Accident and Emergency Departments of hospitals. Suggest why the demand for blood group O is the greatest. (3 marks)

12. 2018Q9a

Although most DNA is stored in the nucleus, mitochondria also have a small amount of their own DNA. The mitochondrial DNA contains some genes coding for enzymes which are essential for oxidative phosphorylation. Mutations of these mitochondrial genes cause Leigh syndrome, which is a rare and fatal metabolic disease.

- a) Explain why gene mutations in the mitochondrial DNA can affect oxidative phosphorylation. (4 marks)

13. 2019Q11 - Essay

Nowadays, keeping pets (such as dogs and cats) is becoming popular in Hong Kong. Some people prefer pure-bred pets to hybrid pets. However, pure-bred pets usually have higher risks of suffering from genetic diseases than hybrid pets because of the ways they are bred. Pure-bred pets are produced by crossing close relatives to keep a pure bloodline. Explain why genetic diseases are often carried by recessive alleles. By comparing the effects of the two breeding processes on the genetic composition of the offspring, discuss why pure-bred pets are at a higher risk of suffering from genetic diseases than hybrid pets. (11 marks)

14. 2020Q9
The following DNA sequence shows the coding strand of part of a gene found in insect species A:
...ATG GTC GTA TAC GCT ACC CTG TCG ATG CTA GCT AGC

- a) Which of the following correctly shows the sequence of the mRNA corresponding to the underlined sequence of this coding strand? Put a ‘✓’ in the appropriate box to indicate your choice. (1 mark)
- ☐ AUG GUC GUA UAC GCU ACC
- ☐ UAC CAG CAU AUG CGA UGG
- b) Using the following codon table, write the amino acid sequence of the protein translated from the mRNA in (a). (2 marks)

		Second base of the codon					
		U	C	A	G		
First base of the codon	U	Phe	Ser	Tyr	Cys	U	Third base of the codon
		Leu		STOP	STOP	C	
	C	Leu	Pro	His	Arg	A	
				Gln		U	
				Asn		C	
						G	
	A	Ile	Thr	Lys	Arg	U	
		Met		Asp		C	
	G	Val	Ala	Glu	Gly	A	
						G	

- c) It was found that the gene has two alleles. The difference between the two alleles in the underlined sequence is highlighted below:
- allele1: ... ATG GTC GTA TAC GCT ACC ...
- allele 2: ... ATG GTC GTA TAC GCT ACC ...
(mutated)
- i) What kind of mutation is this? (1 mark)
- ii) With reference to the codon table, describe how this mutation affects the protein translated from this gene. (3 marks)
- iii) Individuals from insect species A exist in two forms, green bodied and brown bodied. Individuals with a green body have only allele 1. Some of those with a brown body have both allele 1 and allele 2 while others have only allele 2. If the body colour of insect species A was only caused by the mutation of this gene, which allele (1 or 2) would be recessive? Explain your answer. (3 marks)
(Note: Marks will not be awarded for genetic diagrams.)

15. 2021Q6c
Pathogen X is a pathogen that infects humans. Research has discovered an antigen Y present on the surface of pathogen X. Using recombinant DNA technology, antigen Y can be produced and serves as a vaccine to induce immunity against pathogen X.

- c) Refer to the codon table below, answer the questions that follow:

UUU	Phe	UCU	Ser	UAU	Tyr	UGU	Cys
UUC		UCC		UAC		UGC	
UUA	Leu	UCA	STOP codon	UAA	STOP codon	UGA	STOP codon
UUG		UCG		UAG		UGG	
CUU	Leu	CCU	Pro	CAU	His	CGU	Arg
CUC		CCC		CAC		CGC	
CUA		CCA		CAA	Gln	CGA	
CUG		CCG		CAG		CGG	
AUU	Ile	ACU	Thr	AAU	Asn	AGU	Ser
AUC		ACC		AAC		AGC	
AUA	Met	ACA		AAA	Lys	AGA	Arg
AUG		ACG		AAG		AGG	
GUU	Val	GCU	Ala	GAU	Asp	GGU	Gly
GUC		GCC		GAC		GGC	
GUA		GCA		GAA	Glu	GGA	
GUG		GCG		GAG		GGG	

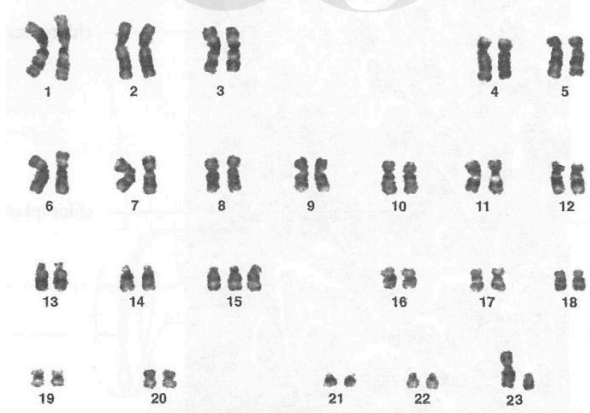
- i) The starting sequence of the coding strand of the gene which encodes antigen Y is shown below: (2 marks)
ATG GCC ATA AAT TGC TGT ...
Referring to the codon table, write the corresponding amino acid sequence of the coding strand shown above.
- ii) Over the years, mutation has occurred in the gene encoding antigen Y in different strains of pathogen X. The variations in the starting sequence of this gene are shown below: (4 marks)
original strain: ATG GCC ATA AAT TGC TGT ...
strain P: ATG GCC ATA AAT TGC TGC ...
strain Q: ATG GCC ATA AAT TGA TGT ...
strain R: ATG GCT ATA AAC TGC TGT ...
One of these strains has the ability to infect people who have been injected with the vaccine containing antigen Y. With reference to the codon table, which strain (P, Q, or R) will that be? Explain your answer.

16. 2021Q10
- In humans, breast milk provides not only nutrients but also protective effects to infants. Recently, scientists discovered a new constituent of breast milk: short RNA fragments enclosed in vesicles. Scientists have very diverse views about the roles of these short RNA fragments. The following are two of the hypotheses:
Hypothesis 1: The short RNA fragments serve as food particles
Hypothesis 2: The short RNA fragments regulate gene expression in infants.
- a) To test Hypothesis 1, scientists performed an experiment of *in vitro* digestion of breast milk. The method is shown below:
- | Method of <i>in vitro</i> digestion with 20 mL of fresh breast milk | |
|---|---|
| Step 1 | Addition of hydrochloric acid solution |
| Step 2 | Addition of enzyme mixture 1 |
| Step 3 | Incubation at 37°C for 20 minutes |
| Step 4 | Addition of sodium hydrogen carbonate solution |
| Step 5 | Addition of enzyme mixture 2 |
| Step 6 | Incubation at 37°C for 30 minutes |
| Step 7 | Incubation at 85°C for 3 minutes |
| Step 8 | Measurement of the level of short RNA fragments and nucleotides |
- i) With reference to the digestion in the human body, what is the importance of Step 1 and Step 4 to the experimental design of this *in vitro* experiment? (3 marks)

ii) What is the purpose of Step 7? (1 mark)

iii) After the *in vitro* digestion, the level of short RNA fragments in the reaction mixture was similar to that of fresh breast milk and no nucleotides were detected. Explain why the results disprove Hypothesis 1. (2 marks)
- b) Scientists will ask scientific questions when designing experiments to test Hypothesis 2. (1 mark)
Suggest **one** example of these scientific questions.

17. 2022Q4a, b
- The photomicrograph below shows the karyotype of a patient who is suffering from a certain brain disease:



- a) What is the gender of this patient? Describe **one** observable feature from the karyotype to support your answer. (2 marks)
- b) i) Describe the abnormality shown in the karyotype. (1 mark)
- ii) State the type of mutation involved in this abnormality. (1 mark)
- iii) How would this abnormality affect the mRNA level in the brain cells of this patient? (1 mark)

18. 2022Q9b
- Antibiotics have been widely used to treat bacterial infections. They work by killing bacteria or inhibiting their growth. However, some strains of bacteria have developed resistance to antibiotics.
- b) Bacterium R is a pathogenic bacterium which possesses an antibiotic resistance gene. Its gene product can break down antibiotic X. Scientists have suggested a new approach to fight against bacterium R. This new approach involves the use of a synthetic polynucleotide which binds to the mRNA transcribed from the antibiotic resistance gene. The expression of the gene is then inhibited. By administering this synthetic polynucleotide together with antibiotic X, bacterium R can be killed.

The base sequences of the synthetic polynucleotide and part of the mRNA are shown below:
Synthetic polynucleotide: AGT GAC TCG GTC AGC
mRNA: ... AUG UCU GUU CCA UCA UCA CUG AGC CAG UCG GCC AUU AAU GCC AAC UAG ...

i) On the mRNA, underline the base sequence to which the synthetic polynucleotide will bind. (1 mark)

ii) Explain how the synthetic polynucleotide can inhibit the expression of the antibiotic resistance gene. (3 marks)

iii) Suggest **one** advantage of using synthetic polynucleotides to fight against antibiotic resistant bacteria. (1 mark)

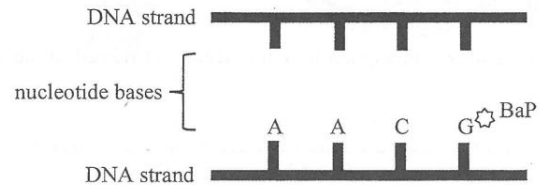
19. 2023Q6b, c
- Colour blindness is an inherited disorder due to defective functioning of the cone cells in the retina. There are many types of colour blindness. For example, people with red-green colour blindness fail to distinguish between red and green colors while those with total colour blindness experience total loss of colour vision.
- b) Red-green colour blindness is caused by a recessive allele on the X-chromosome while total colour blindness is caused by a recessive allele which is located on an autosome. The table below shows the percentage occurrence of red-green colour blindness and total colour blindness in men and women:

	Men	Women
Red-green colour blindness	8%	0.5%
Total colour blindness	0.00001%	0.00001%

- With reference to the inheritance of the two types of colour blindness, suggest why the occurrence of red-green colour blindness in men as compared to women differs from that of total colour blindness. (4 marks)
- c) The pedigree shows the inheritance of red-green blindness in a family:
-
- i) The couple is expecting another child. Using 'B' to represent the allele for normal vision and 'b' to represent the allele for red-green colour blindness, construct a genetic diagram to find out the probability of this newborn being a girl with red-green colour blindness. (Note: Punnett square is not accepted.) (4 marks)
- ii) Individuals 4 and 5 are twins. Can you determine whether they are identical twins or fraternal twins? Explain your answer. (2 marks)

20. 2024Q2

BaP is a carcinogenic chemical which is commonly found in grilled meats. It can attach randomly to the nucleotides of DNA molecules. When it is attached to guanine (G), this G will be misread as thymine (T). The diagram below shows part of the nucleotide sequence of one strand of a DNA molecule with BaP attached to a G:

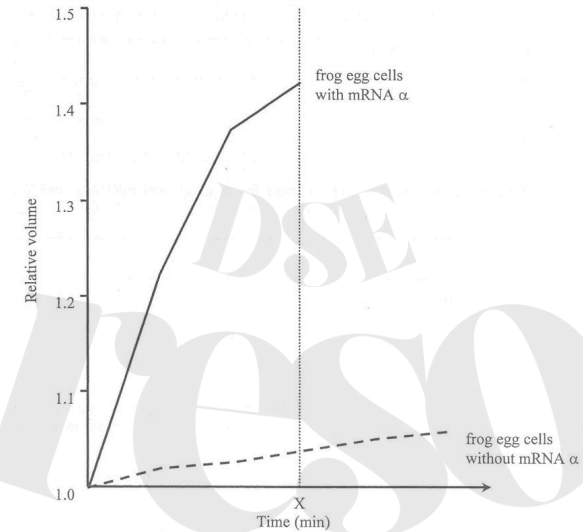


- On the above diagram, write down the nucleotide sequence found in the opposite strand of the DNA when misreading happens. (1 mark)
- Suggest **one** reason why this type of mutation may **not** affect the functioning of the protein formed. (1 mark)
- If this type of mutation accumulates over time in the DNA molecules, there is a chance that it will affect the functioning of the protein formed and subsequently lead to tumor formation. Suggest which cellular process this protein controls. (1 mark)

21. 2024Q5

In an experiment, mRNA α was isolated from a mammalian cell and then injected into a frog egg cell. The expression of mRNA α eventually led to the presence of protein α on the cell membrane of the frog egg.

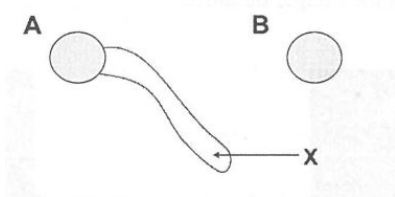
- Describe how the injected mRNA α led to the presence of protein α on the cell membrane of the frog egg. (3 marks)
- In another experiment, frog egg cells received an injection of a fixed amount of water with or without mRNA α . After that, these two types of frog egg cells were transferred to pure water. The changes in the relative volumes ($\frac{\text{new volume}}{\text{original volume}}$) of these two types of frog egg cells are shown in the graph below:



- Explain why there was an increase in the relative volume of the frog egg cells without mRNA α after they were transferred to pure water. (2 marks)
- Based on the difference shown in the results of the two types of frog egg cells, deduce the function of protein α on the cell membrane. (3 marks)
- Suggest why no data were obtained from frog egg cells with mRNA α after X minutes. (1 mark)

22. 2024Q10a, c

In an investigation, pollen grains collected from a single flower were cultured in an artificial medium. After 48 hours incubation, they were observed under a light microscope. Two types of pollen grains with different appearances were observed, as shown in the diagram below:



- a) The number of each type of pollen grain is approximately the same. It is known that the formation of structure X is controlled by a single gene. Deduce the genotype of the parent plant producing these two types of pollen grains. (4 marks)
- b) If these two types of pollen grains land on a stigma of the flower of the same species, which type of pollen grains will lead to formation of seed? Explain your answer. (3 marks)
- c) 100 seeds were collected from the parent plant in (a) after self-pollination. According to your answer in (b), complete the following table to show the proportion of genotypes in these seeds. (1 mark)

Genotype	Homozygous dominant	Heterozygous	Homozygous recessive
Proportion (%)			

Ans

2012Q15: B (63%)	2012Q18: D (78%)	2012Q19: A (31%)	2013Q4: A (81%)
2013Q11: C (35%)	2013Q12: C (80%)	2013Q13: B (84%)	2013Q15: B (47%)
2013Q16: D (55%)	2013Q17: B (71%)	2013Q19: B (59%)	2014Q10: A (50%)
2014Q11: A (31%)	2014Q12: B (80%)	2014Q16: A (71%)	2015Q16: C (88%)
2015Q26: A (42%)	2016Q4: B (33%)	2017Q8: B (79%)	2017Q9: A (53%)
2018Q12: A (39%)	2018Q13: C (77%)	2018Q28: A (63%)	2019Q13: D (45%)
2019Q14: D (63%)	2020Q18: C (73%)	2020Q19: A (52%)	2020Q33: D (43%)
2020Q34: D (57%)	2020Q35: A (38%)	2020Q36: C (41%)	2021Q1: D (60%)
2021Q2: D (63%)	2021Q3: C (57%)	2021Q11: C (63%)	2021Q12: C (58%)
2021Q13: A (53%)	2021Q14: A (73%)	2021Q15: D (75%)	2021Q16: D (40%)
2021Q17: C (38%)	2022Q2: C (60%)	2022Q3: A (35%)	2022Q6: D (51%)
2023Q4: A (42%)	2023Q24: D (32%)	2023Q27: B (39%)	2023Q28: A (54%)
2023Q30: C (64%)	2024Q2: A (28%)	2024Q5: A (25%)	2024Q6: C (41%)
2024Q16: C (64%)	2024Q17: B (56%)	2024Q18: C (63%)	

- 2012Q8b, c
- 8 (b) (i)
 - as both parents are pure-bred plants, the F₁ produced are all heterozygous (1)
 - in heterozygous condition, dominant alleles will be expressed while recessive alleles will be masked (1)
 - therefore, purple colour and smooth surface are the dominant phenotypes (1)
- (ii)
 - only one type of gametes will be produced from the pre-bred corn plant with yellow and wrinkled kernels (1)
 - as the individuals in F₁ are heterozygous with genes for the two traits located on different chromosomes, the alleles for the traits are assorted independently (1)
 - therefore, four different types of gametes will be produced from the F₁ (1)
 - after random fertilisation, zygotes with four different combinations of genotypes will be produced (1)
 - leading to the expressions of all the possible phenotypes (A, B, C and D) in the F₂ generation (1)
- (c)
 - Mendel collected a large amount of experimental evidence (1)
 - and based on his observation on the number of offspring with different phenotypes (1)
 - he worked out a possible explanation for the observation by logical deduction (1)

2013Q4

- 4 (a) - as Mary's father is red-green colour blind, he must have a X chromosome bearing the recessive allele for red-green colour blindness (1)
 - being a female, Mary must have inherited the X chromosome bearing the recessive allele from her father (1)
 - on the other hand, Mary is normal, therefore she must have a X chromosome bearing dominant allele for normal eye sight (1)
 - therefore, Mary is a heterozygote (1) (4)
- (b) (i) No, it is because
 - the photomicrograph cannot reveal the genetic make-up of the foetus / the photomicrograph only shows the number or appearance of chromosomes (1)
 - whereas red-green colour blindness is due to the presence of a certain allele (1) (2)
- (ii) - the foetus is a female (1) because
 - the 23rd pair of chromosomes in the karyotype have similar size / length (1)
 - and thus the sex chromosomes of the foetus are both X chromosomes (1) (3)

9 marks

2014Q6a

- 6 (a) • dogs have many more copies of gene A in the genome than wolves (1)
 • these gene copies will be transcribed into mRNA which, in turn, translated into amylase (1)
 • more amylase will be produced in dogs (1), resulting in higher amylase activity (3)

2015Q2b

- 2 (b) • stage A (1)
 • genetic materials are dispersed / loosely packed at stage A (1), which indicates that the DNA molecules are ready for transcription (2)

2015Q4

- 4 (a) blood type O (1) (1)
- (b) (i) ii (1) (1)
- (ii) Father: I^Ai (1) Mother: I^Bi (1) (2)
- (c) • antibodies A and B are already present in Roger's blood (1) (1)
 • these antibodies will act against the antigens of the parents' red blood cells (1) (1)
 • and causes blood cells clumping / hemolysis of blood cells (1) if Roger receive blood transfusion from his parents (1) (1)

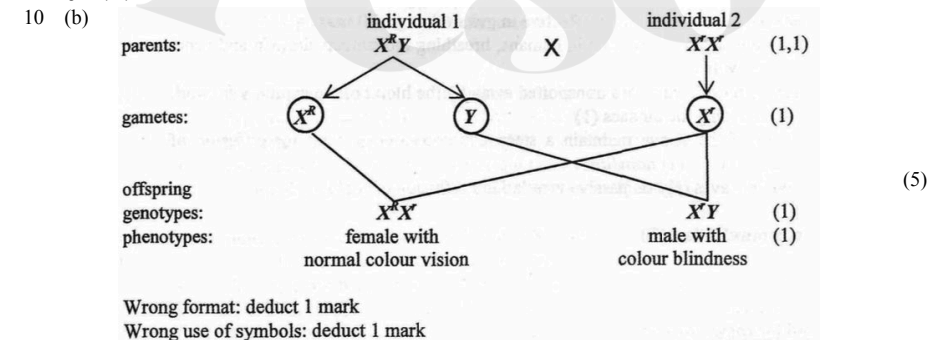
7 marks

2016Q8

- 8 (a) • those with the cap removed could regenerate the cap / the stalk and foot alone could regenerate the cap (1)
 • those with the foot removed could not regenerate the foot / the cap and stalk alone could not regenerate the foot (1) (2)
- (b) (i) (1) RNA (1) (1)
- (2) • RNA directs the protein synthesis through translation (1)
 • the protein produced determine the morphology of the cap by acting as enzymes or structural proteins (1) (2)
- (ii) • the final morphological feature of the cap resembles that of Species 2 (1)
 • showing that the trait is determined by the foot of Species 2 but not the stalk from Species 1 (1)
 thus the permanent heredity information is stably stored in the foot (2)
- (c) • scientific knowledge is based on or derived from observations of the natural world (i.e. empirically based or evidence based) (1)
 • Hammerling's observation on the regeneration of the cap from the algae leads to the confirmation of the location of genetic materials (1)
- Or
- doing science requires creativity and imagination (1)
 • Hammerling designed experiment to test his hypothesis, the process required creativity and imagination (1) (2)
- Or
- science is a process of ongoing inquiries (1)
 • Hammerling's grafting experiment has led to the inquiries about short-lived instructions (1)

9 marks

2016Q10b, c, d



- (c) (1)
- (d) • as David is a male, he has only one X-chromosome (1)
 • having normal colour vision, this X-chromosome must bear the allele for normal colour vision (1)
 • and this X-chromosome that bears the allele for normal colour vision must be inherited to his daughter (1)
 • because the allele for normal colour vision is dominant over the allele for colour blindness / will be expressed over the allele for colour blindness (1)
 • even if the baby girl receives an allele for colour blindness from Daisy, the baby girl will have normal colour vision (1) (5)

2017Q4c

- 4 (c) • B carries the genetic materials (1) coding for the hormone / digestive enzymes
 • transcription takes place in B to produce mRNA (1)
 • which is transported to A for translation / protein synthesis (1)
 • to produce protein hormones / digestive enzymes (1) for discharge / secretions (4)

2017Q10

- 10 (a) (i) • discontinuous trait (1)
 • because there are distinctive categories with no intermediate categories (1) (2)

- (ii) • in case I, non-rolling offspring appeared even when both parents were tongue rollers (1)
 • this shows that the allele for non-tongue rolling was masked in the parents (1) hence, tongue rolling should be the dominant phenotype and non-tongue rolling should be the recessive phenotype

OR

- in case I, non-tongue rolling offspring must have inherited at least one allele for non-tongue rolling from at least one of their roller parents (1)
 • however, both parents showed tongue rolling phenotype, i.e. non-tongue rolling phenotype was masked (1) hence, tongue rolling should be the dominant phenotype and non-tongue rolling should be the recessive phenotype (2)

- (b) • no, it did not support the conclusion (1)
 • if non-tongue rolling was the recessive phenotype, all the offspring of the non-tongue rolling parents would be non-tongue rollers (1) (2)

- (c) (i) • the genetic composition of the identical twins is exactly the same (1)
 • they should show the same phenotype (1) if the trait is controlled by genetic factors (2)

- (ii)

Conclusion	Evidence
Genetic factor plays a significant role in the determination of the tongue rolling trait.	82% of the identical twins showed same phenotype (1)
There are other factors influencing the tongue rolling trait.	18% of the identical twins showed different phenotypes (1)

 (2)

- (d) (i)

Ideas about Science	
Science is a process of ongoing inquiries.	✓
Science is affected by social and cultural factors.	
Scientists having the same set of data may not arrive at the same conclusions.	
Scientific investigation need not necessarily be doing experiments in laboratories	✓

 (2)

- (ii) • scientists who conducted tongue rolling experiments in 1965 or 1971 should have been skeptical and have looked for evidence from different sources / perspectives (1) / have been ready to review / challenge current ideas / knowledge / theories (1)

13 marks

2018Q5c

- 5 (c) • mother will produce only one type of eggs with an X chromosome (1)
 • father will produce two types of sperm cells, one with an X chromosome while another one with a Y chromosome (1)
 • the sex of offspring will be determined by which type of sperm cells is involved in the fertilisation (1), which is a random process (3)

2018Q6a

- 6 (a) • the blood cells of blood group O contain neither antigen A nor B (1)
 • therefore, it can be transfused to patients of any of the ABO blood groups / group O individuals are universal donors (1)
 • it will be used for transfusion to other blood groups if there are not enough blood reserve for other blood groups, (1)
 • the population of blood group O individuals is the largest of all blood groups in Hong Kong (1) (max. 3)

2018Q9a

- 9 (a) • gene mutation involves alteration in the nucleotide / base sequence / triplet code in DNA of a single gene (1)
 • polypeptide produced from the mutated gene has a different amino acid sequence (1)
 • it folds into an enzyme with a different active site shape / 3D conformation (1)
 • this enzyme can no longer function in the oxidative phosphorylation (1) / fail to produce a functional enzyme / active site of this enzyme can no longer fit the substrate (4)

2019Q11

- 11 **Concept on genetic diseases (max. 3 marks)**
 spontaneous mutations may give rise to dominant or recessive alleles, some of these mutations may be beneficial to the individuals, some may be disadvantageous,
 • if genetic diseases are encoded in the form of dominant alleles, the diseases will be expressed in both homozygous and heterozygous forms (1), therefore they will be removed by natural selection more readily (1), so the chance of passing such dominant alleles to the next generation is lower (1) (max. 3)
 • if genetic diseases are encoded in the form of recessive alleles, the diseases will be expressed only in homozygous recessive condition (1), therefore the chance of removal by natural selection is lower (1), the recessive allele will be passed to the next generation by carriers in heterozygous condition (1)

Effect of breeding methods on genetic compositions (max. 5 marks)

- the parents of pure-bred pets are from among close relatives, i.e. they share the same gene pool / have similar genetic composition (1), as a result, the genetic composition of their offspring will be very similar to each other (1)
 • on the other hand, parents of hybrid pets are from different gene pools / have very different genetic compositions (1), as a result, the genetic composition of their offspring will be different from each other (1)
 • after several generations, the pure-bred offspring have a higher chance of being homozygous in various loci (1), whereas the hybrid offspring have a higher chance of being heterozygous in various loci (1) (max.5)
 • therefore, it is more likely for pure-bred pets to develop genetic diseases, which are only expressed in homozygous recessive or double recessive conditions (1)
 • and it is less likely for hybrid pets to develop genetic diseases, as recessive alleles for genetic diseases are masked by their dominant counterparts in heterozygous condition (1), i.e. no genetic diseases are expressed

Effective communication (0-3)max. 3
11 marks

2020Q9

- 9 (a) AUG GUC GUA UAC GCU ACC (1)
- (b) Met - Val - Val - Tyr - Ala - Thr (2)(deduct 1 mark for each mistake)
If candidates have chosen wrong answer in (a), accept:
Tyr - Gln - His - Met - Arg - Trp (2)
- (c) (i) gene mutation (1)
- (ii) • the mutation results in a stop codon instead of tyr (1)
• as result, the protein chain produced will be much shorter than the original one / the sequence after the new stop codon will not be translated / the protein chain produced will have a different amino acid sequence from that of the original one (1)
• therefore, the protein formed will have a different shape (1) (3)
- If candidates have chosen wrong answer in (a), accept:
• the mutation results in a different amino acid (1)
• as result, the protein chain produced will have a different amino acid sequence from that of the original one (1)
• Therefore, the protein formed will have a different shape (1)
- (iii) • allele 1 is the recessive allele (1)
• individuals with green body contain only allele 1, this shows that allele 1 codes for green colour (1)
• some individuals with brown body colour contain both alleles (1 and 2), showing that the effect of allele 1 is masked by allele 2 (1)
- OR
- the phenotype shown in the heterozygous condition is the dominant phenotype (1)
• therefore brown body colour is dominant over green body colour / green body colour is recessive (1)
• therefore, allele 1 carried by individuals with green body is recessive (1)

10 marks

2021Q6c

- 6 (c) (i) • Met Ala Ile Asn Cys Cys (2 marks, deduct 1 mark for each mistake) (2)
- (ii)

Concept for mark award
• correct choice of strain (1)
• the effect of the stop codon on the polypeptide produced (1)
• the effect on the antigen produced (1)
• the failure to recognise the pathogen (1)

 (4)
- e.g.
• strain Q (1)
• TGA becomes the stop codon and so the polypeptide produced is much shorter (1)
• therefore, no antigen Y will be produced / the antigen produced will have a different shape (1)
• the memory cells for antigen Y can no longer recognise the pathogen (1)

2021Q10

- 10 (a) (i) • Step 1 simulates / mimics the pH condition / acidic condition / the presence of gastric juice in the stomach (1)
• while Step 4 simulates / mimics the pH condition / alkaline condition / the presence of pancreatic juice / intestinal juice / bile juice / neutralisation of the gastric juice in the small intestine (1) (3)
• and thus provides optimal / proper / suitable pH conditions for enzymatic digestion (1)
- (ii) • to stop the activity of the enzymes / the enzymatic reaction / denature the enzymes (1) (1)
- (iii) • this shows that the short RNA fragments were not digested into nucleotides (1)
• the RNA fragments are too large to pass through the wall of the small intestine / the small intestine cannot absorb the short RNA fragments as they are too large (1) to pass through effectively (2)
- (b) any **one** of the following:
• check whether the cells can pick up these short RNA fragments (1)
• check if these short RNA fragments in the milk vesicles can be detected in the infants, other than in the digestive tract in vivo, e.g. other organs / in the blood (1) (1)
• check if the presence of these short RNA fragments will affect gene expression (1)
(accept other reasonable answers)

7 marks

2022Q4a, b

- 4 (a) • male (1)
- Concept for mark award
• relating the features of sex chromosomes (i.e. shorter Y chromosome) to sex determination (1)
- e.g.
• because chromosomes in the 23rd pair are of different lengths or sizes, showing that the pair is composed of an X chromosome and a Y chromosome / the shorter chromosome in the 23rd pair or the last pair is a Y chromosome (1), therefore, the patient is a male
- (b) (i) • an extra copy of chromosome 15 (1) (1)
- (ii) • chromosomal mutation (1) (1)
- (iii) • level of expression of mRNA of the genes located on the 15th set of chromosomes is increased (1) as there are three chromosomes (1)

2022Q9b

9 (b) (i) • UCA CUG AGC CAG UCG (1) (1)

- (ii) Concept for mark award (3)
- role of mRNA (1)
 - role of tRNA / ribosome (1)
 - failure of translation (1)

e.g.

- after binding, the single-stranded template is no longer available / the binding region becomes double stranded (1)
- the tRNA cannot bind to the template to transfer the corresponding amino acid / ribosome cannot bind to the template (1)
- translation cannot proceed (1) at the binding region

- (ii) any **one** of the followings: (1)
- can design a new synthetic polynucleotide any time once we know the sequence of antibiotic resistance gene (1)
 - the action is very specific (1) as it targets the resistance gene

2023Q6b, c

- 6 (b) Concept for mark award (4)
- correct comparison of the occurrence of red-green colour blindness in men and women with that of total colour blindness in men and women (1)
 - relate X-linked inheritance to the greater chance of men suffering from red-green colour blindness (2)
 - relate autosomal inheritance to the same chance of suffering from total colour blindness in both sexes (1)

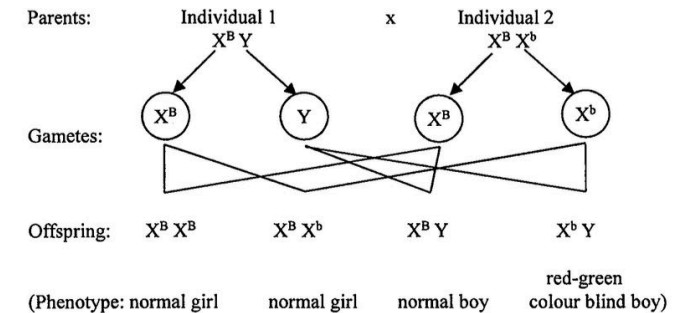
e.g.

- the percentage of men suffering from red-green colour blindness is greater than that of women, while the percentage of men suffering from total colour blindness is the same as that of women (1)
- the allele for red-green colour blindness is located on the X-chromosome, since men have only one X-chromosome, they will have the disorder once they receive an X-chromosome carrying the recessive allele / allele for red-green colour blindness (1)
- women have two X-chromosome, so they will have the disorder only if they have recessive alleles on both X-chromosomes / the effect of the recessive allele will be masked by the effect of the dominant allele (1), as a result, women have a lower chance of having red-green colour blindness
- the allele for total colour blindness is located on an autosome, so both men and women have an equal chance of inheriting the recessive allele / allele for total colour blindness (1)

- 6 (c) (i) Concept for mark award (4)
- format (1)
 - genotypes of parents (1)
 - gametes and genotype combinations of offspring (1)
 - correct probability (1)

e.g.

Let X^B be the X chromosome carrying the allele for normal vision
 X^b be the X chromosome carrying the allele for red-green colour blindness
 Y be the Y chromosome



the probability of this newborn being a girl with colour blindness is zero

- (ii) • cannot be determined (1) (2)
- as there is still a chance that they develop from two eggs with X chromosomes carrying the mutation / only shows the inheritance of one trait, cannot tell if they have the same or different phenotypes in other traits (1)

2024Q2

	Concept for mark award	Example	Marks
2(a)	correct complementary base pairing	TTGA	(1)
2(b)	Some triplet codes encode the same amino acids / conformation of protein	any one of the following: - the new triplet code encodes the same amino acid as the original amino acid / different triplet codes may code for the same amino acid - the new amino acid is located in a region which does not affect the functional part of the protein / the active site of the enzyme	(1)
2(c)	cellular activities related cancer formation	any one of the following: - control of cell division / mitotic cell division - control of cell growth - control of cell death - control of cell cycle - control of cell differentiation - DNA repair	(1)

3 marks

2024Q5

	Concept for mark award	Example	Marks
5(a)	translation process	the mRNA α is translated at rough endoplasmic reticulum / ribosome to form a polypeptide chain	(1)
	folding of polypeptide to form protein	the polypeptide chain will fold into protein α / a protein with a specific shape	(1)
	transport to membrane	protein α / the protein formed will be transported to the membrane	(1)
5(b)(i)	comparison between cell sap and bathing solution in water potential	pure water has a higher water potential than that of the cytoplasm / cell sap in the frog egg cell	(1)
	correct description of water movement	there is a net movement of water into the frog egg cell by osmosis / water moves from a region of higher water potential to a region of lower water potential, i.e. from water into the frog egg cell	(1)
5(b)(ii)	comparison of the results from normal frog egg cell and frog egg cell injected with mRNA a	the relative volume of the frog egg cell injected with mRNA α increases much faster than that injected with water only / at the time point X, the relative volume of the frog egg with mRNA a is much higher than that injected with water only	(1)
	inference from the difference	this shows that the presence of protein α has increased the permeability of the membrane to water	(1)
	deduction	therefore, protein α is probably a channel protein for the passage of water	(1)
5(b)(iii)	correct explanation	the frog egg cells injected with mRNA α have burst	(1)

9 marks

2024Q10a, c

	Concept for mark award	Example	Marks
10(a)	recognise that pollen grains are formed after meiotic cell division / haploid	pollen grains are formed after meiotic cell division / haploid, they carry one set of chromosomes only	(1)
	deduction from phenotypes to genotypes	structure X is developed in pollen grain A, thus the allele carried by pollen grain A should be responsible for this trait / character / characteristic / the development of structure X	(1)
	deduction about the genotype of parent plant	structure X failed to develop in pollen grain B, thus the allele carried by pollen grain B should be responsible for this trait / character / characteristic	(1)
10(c)	correct percentage	therefore, the parent plant that produced these two types of pollen grain should have possessed both alleles, i.e. heterozygous homozygous dominant: 50%; heterozygous: 50%; homozygous recessive: 0% or homozygous dominant: 0%; heterozygous: 50%; homozygous recessive: 50%	(1)